The Gender Pay Gap and Its Determinants Across the Human Capital Distribution

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Abstract

This paper links American Community Survey data and postsecondary transcript records to examine how the gender pay gap varies across the distribution of education credentials for a sample of 2003-2013 graduates. Although recent literature emphasizes gender inequality among the most-educated, we find a smaller gender pay gap at higher education levels. Field-of-degree and occupation effects explain most of the gap among top bachelor's graduates, while work hours and unobserved channels matter more for less-competitive bachelor's, associate, and certificate graduates. We develop a novel decomposition of the child penalty to examine the role of children in explaining these results.

Keyword*:* gender pay gap, postsecondary transcript records, college selectivity, human capital, field and occupation choice, labor supply, child penalty

JEL Classification: I24, I26, J16, J31

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I. Introduction

In 2022, the average woman in the U.S. workforce earned 72.3 cents for every dollar earned by the average man.[1] Canonical research has shown that the average gender pay gap cannot be explained by gender differences in educational attainment and other forms of human capital (Altonji and Blank, 1999; Bertrand, 2011; and Blau and Kahn, 2017). More recent research has studied gender inequality among graduates from top universities or in occupations that require advanced degrees and specialized skills (Noonan et al. 2005; Bertrand et al. 2010; Matsa and Miller 2011; Goldin 2014; Bursztyn et al. 2017; Fernandez and Campero 2017; Wiswall and Zafar 2018; Houghton 2020; Cortés et al. 2021; Foster et al. 2023; Wasserman 2023). The traditional focus on the average and the newer focus on the elite have generated important findings, but it is unclear how well these findings apply to an increasingly stratified and socioeconomically diverse U.S. population. We take up that question in this paper by studying how the gender pay gap varies across a wide distribution of education credentials.

Rising U.S. inequality and diversification of the higher education system (Gerber and Cheung 2008; Hoxby 2009; Lovenheim and Smith 2022) suggests a need to move beyond the average gender pay gap to an analysis of how the gap varies across different subpopulations. On the one hand, high returns to specialized professional skills (Deming and Kahn 2018; Deming and Noray 2020), high returns to intensive hours in certain professions (Goldin 2014), and the male-dominated networks that often persist in these environments, may widen gender inequality at the top of the human capital distribution. On the other hand, exposure to unpredictable shift-scheduling

[1] Authors' computations from public-use American Community Survey data on individuals aged 25-54 and with positive earnings. Among individuals employed for at least 50 weeks during the year, the corresponding figure is 74.5 cents on the dollar.

(Schneider and Harknett 2019), lack of employer-provided work-family support, and tenuous childcare arrangements and social-support networks (Su and Dunifon 2017; Harknett et al. 2020; Cortés and Pan 2023) may widen gender inequality at the lower end of the distribution (Binder 2021). While some research has investigated the relationship between prevailing wage inequality and the average gender pay gap (e.g. Blau and Kahn 1997, 2000; Mulligan and Rubinstein 2008; Fortin et al. 2017), little work to our knowledge examines gender inequality across an unequal distribution of skills and opportunities.

We develop a dataset that is uniquely qualified for such an investigation by linking 2005-2019 American Community Survey (ACS) respondents to their postsecondary transcript records, drawing on a partnership between the U.S. Census Bureau and higher education systems covering most postsecondary graduates in 8 different states. We construct a sample of 2003-2013 graduates, whose credentials range from short-term certificates to bachelor's degrees conferred by highly-selective colleges, and for whom we can observe labor-market outcomes for up to 16 years following graduation. While our dataset is not a random sample of the population, we show that it differs only slightly from representative datasets of young graduates drawn from the ACS and the National Survey of College Graduates on key characteristics. Our dataset strikes us as exceptional in its combination of broad representativeness and fitness for detailed subgroup analysis.

Our investigation generates two novel findings regarding contemporary gender inequality. First, despite the recent focus of the literature, we find the gender pay gap is not uniquely large among the elite. By one measure, we find that the gap *falls* as degree level rises; by another, the gap is constant across degree levels. Second, the proximate mechanisms determining the pay gap vary considerably across degree levels. For example, while gender differences in field of degree explain a substantial portion of the gap at the highest levels, they are much less important at lower

levels. Controls for field, occupation, and work hours explain almost the entire gap at the top, but they leave a substantial unexplained portion at other levels. We largely confirm these findings in subsample analyses of full-year-employed workers and bachelor's graduates who majored in Business or Economics.

We then extend these findings by analyzing the role of parenthood exposure in explaining them, drawing on administrative birth-record data that is available for a subset of our cohorts. We estimate a similar (and large) effect of children on the pay gap at certificate and top bachelor's levels. However, observed controls explain only about half of this effect at the certificate level. That is, the "child penalty" operates primarily through unobserved channels for these graduates; in contrast, it operates almost entirely through occupation and work-hours effects for top bachelor's graduates. Accordingly, while certificate and top bachelor's graduates differ substantially in their fertility timing, this difference does not account for the observed variance across degree levels in the mechanisms driving the gender pay gap.

Our results indicate that different mechanisms of gender pay inequity are important at different positions in the labor market for recent graduates. While Roy-style models of career selection may generate most of the observed pay disparity among the most-educated, frictional models of work-family constraints, search, or discrimination may play an important role among the less-educated. These findings contextualize the recent focus of the literature on the most-educated segments of the population and indicates a need for greater focus on less-educated segments, where the pay gap is at least as large and less well-explained.

Our paper also extends literature on the contribution of field choice to the pay gap (e.g. Altonji 1993; Brown and Corcoran 1997; Altonji et al. 2012; Black et al. 2008; Bertrand 2018; Francesconi and Parey 2018; Sloane et al. 2021; Humphries et al. 2024) by revealing important heterogeneity

that is masked in studies of the general college-graduate population.[2] It also connects standard pay-gap decomposition techniques to an analysis of the child penalty, which continues a trend in that literature to estimate the penalty's underlying mechanisms and heterogeneity across populations (e.g. Hotz et al. 2018; Jee et al. 2019; Kleven et al. 2019a, 2019b; Sandler and Szembrot 2019; Kleven et al. 2023).

The paper proceeds as follows. In the next section, we describe the construction of our dataset and some basic summary statistics. Section III describes our decomposition methodology. Section IV presents our main gender pay gap decompositions. Section V presents our child penalty analyses. Section VI offers concluding remarks.

II. Data

We source postsecondary transcript records from a subset of the state higher education systems that participate in the Post-Secondary Employment Outcomes (PSEO) partnership:[3] Connecticut State Colleges and Universities, the City University of New York, the Louisiana Board of Regents, the Missouri Department of Higher Education and Workforce Development, the Ohio Department of Higher Education, the State Council of Higher Education for Virginia (SCHEV), the State University of New York, the Texas Higher Education Coordinating Board, and the Utah System of Higher Education. These systems cover the vast majority of graduates from publicly controlled

[2] Comparison of our full-sample results to those of Brown and Corcoran, presented in our working paper (Binder et al. 2023), suggests that field-of-degree sorting remains an important determinant of the gender pay gap among today's baccalaureate graduates, but its effect size has declined substantively with time. This observation matches the multi-cohort studies of Bertrand (2018) and Sloane et al. (2021).

[3] For more information about the PSEO data, go to: https://lehd.ces.census.gov/data/pseo_experimental.html

postsecondary institutions in the given states. SCHEV also covers most private schools in the state of Virginia. As shown in Appendix Table A1, our coverage rate of all postsecondary graduates ranges from 44% in New York to 87% in Virginia, with a mean of 65%.

The transcripts we observe are person-degree[4] records that include information on institution, degree type (short-term certificate, 1-2-year certificate, associate, or bachelor's), degree field, and date of graduation. Each graduate is assigned a Protected Identification Key (PIK) by the U.S. Census Bureau. These PIKs allow the transcript data to be matched with person-level responses to the American Community Survey (ACS). We observe annual transcript data from most systems beginning in the early 2000s, and ACS data for survey years 2005-2019.

A. Main Sample

We use this information to construct a dataset based on the following criteria: individuals who graduated in 2003-2013, who were between the ages of 19 and 42 at graduation, who were between the ages of 19 and 59 when they were sampled by the ACS, and who graduated prior to being sampled by the ACS. Each graduating cohort can be observed for a minimum of six and maximum of 16 years following graduation. As recorded in Appendix Table A1, our sample contains approximately 630,000 graduates, nearly 30% of which graduated from a Texas institution,[5] 23% of which graduated from a New York institution, 17% of which graduated from an Ohio institution, and 15% of which graduated from a Virginia institution.

[4] If an individual earns multiple degrees, there is an observation in the data for each degree earned. We have verified that negligible shares of our sample appear in the PSEO data as both a sub-baccalaureate degree earner and a baccalaureate degree earner.

[5] Our Texas sample covers graduates from all 110 public institutions in the state, amounting to 77% of all graduates.

Table 1 presents summary statistics. Women comprise nearly 59% of observed graduates, reflecting known gender differences in degree attainment in the United States and other advanced countries. Most individuals are in their 30s. The average individual graduated in 2008 and is observed six years later, during the 2014 ACS data year. Most transcripts list a bachelor's degree, while 31 percent list an associate degree and the remaining 9 percent list a certificate.

Table 1 also records summary information on annual earnings. The average woman in our sample earned $38,700, while the average man earned $53,200, leading to a gender gap of $14,500, or 27.3 percent of the average man's earnings. We find a similar gap of $13,600, or 26.4 percent of the average man's earnings, when we consider wage and salary earnings only.

B. Gauging External Validity

Before proceeding, we provide some context for this dataset by investigating its generalizability to the U.S. population. We do so in two complementary ways. First, we compare key demographic and economic statistics between our samples and population-representative datasets. For our sample of associate degree holders, we draw two different comparison samples from the 2012-2016 ACS.[6] The "target comparison" sample only considers individuals born in the 8 states for which we have transcript data and reweights the sample on age to match the age distribution in our analysis sample. The "national comparison" sample considers all individuals aged 19-44 and does not adjust the baseline ACS weights. Both comparison samples match men's average earnings and labor supply almost exactly (Appendix Table A2), but record slightly lower figures for women: the baseline gender earnings ratio is .706 in our analysis sample, and .645-.650

[6] Because the ACS does not record information on certificate degree receipt, it is impossible to do this validation exercise for our sample of certificate graduates.

in the comparison samples. Conditional on positive wage-and-salary earnings, the ratio is .748 in our analysis sample and .707-.711 in the comparison samples.

For the sample of bachelor's degree holders, we leverage the National Survey of College Graduates (NSCG) to finetune our comparison (Appendix Table A3). Our target comparison sample explicitly considers individuals who earned their first bachelor's degree in the 8 PSEO states from a public institution and reweights the data on years-since-graduation in addition to age. This sample contains a statistically indistinguishable baseline gender earnings ratio (.727 as opposed to .709 in our analysis sample). The national comparison sample, which considers all NSCG respondents aged 22-40, contains a slightly lower gender earnings ratio (.662 as opposed to .709; .713 as opposed to .742 excluding those with zero earnings). Both samples also closely match the analysis sample in terms of education outcomes. For example, 29.5% of the individuals in our analysis sample went on to earn an advanced degree, while 25.9%-29.2% did in the comparison samples. Approximately 20.4% of the analysis sample earned a BA degree in a Business or Economics field, while 19.0%-20.7% did in the comparison samples.

Thus, while our samples differ slightly from population-representative samples of young graduates, the differences rarely approach economic significance—particularly for the target comparison samples. If anything, the fact that our samples slightly understate the gender earnings gap relative to national samples of young graduates lends a bit of conservatism to our results.

As a complementary test of our sample's generalizability, we examine the characteristics of institutions as opposed to individual graduates. Specifically, we compare average characteristics among our baccalaureate institutions to national averages computed from the 2004-05 edition of the U.S. Department of Education's College Scorecard database.[7] Appendix Table A4 considers

[7] College Scorecard | College Scorecard (ed.gov).

two national comparisons: all baccalaureate institutions appearing in the 2004 Barron's Guide, and the subset of these institutions that are public. Institutions in the PSEO sample have similar average characteristics to these comparison institutions, particularly when we limit to public institutions. Mean admissions rates and SAT scores look similar across the groups, while PSEO institutions have very slightly lower instructional expenditures and completion rates. We interpret these statistics as evidence that bachelor's-degree-earning students in the PSEO data graduate from institutions that resemble typical bachelor-degree granting public U.S. institutions.

C. Ranking of Degree Levels

To distinguish among the variety of postsecondary institutions in the United States, we merge 2004 Barron's Competitiveness ratings for bachelor's programs onto our dataset at the institution level. Barron's rates programs with competitive admissions policies (i.e. that are not open enrollment), and applies a 1-7 rating scale to the programs it rates. We recoded this to a 1-6 scale, pooling the two least competitive categories (which were small), and assigned a seventh category to programs left unrated by Barron's. This category is small, accounting for under three percent of the full sample.

Table 2 presents student-weighted average institutional characteristics by Barron's category, along with student and institution counts. The table shows substantial stratification in institutional selectivity, investments, and graduation outcomes. While nearly two-thirds of applicants to Most Competitive institutions in our sample are rejected, only 4.6 percent of applicants to Non-Competitive institutions are rejected. Most Competitive institutions spend over twice as much on instruction as do Less and Non-Competitive institutions. While under 30 percent of students at Non-Competitive institutions graduate in four years, over 90 percent of those at Most Competitive

institutions do. Male graduates from Most Competitive institutions earn 63.2 percent more on average than do male graduates from Non-Competitive institutions.

Thus, in total, we consider ten different postsecondary *degree levels*: short-term certificate, one- or two-year certificate, associate, and seven levels of bachelor's degrees.[8]

D. Subsample to Investigate Effects of Children

We use PIK assignments to merge information from the Census Household Composition Key (CHCK) onto our main sample. The CHCK links each birth from 1997-2021 to a set of parent(s) based on administrative records from the Social Security Administration (see Genadek et al. 2021). The advantage of this dataset over the ACS is that the latter can only identify parent-child linkages based on the relationship to household head variable. This makes it difficult to associate parents with children in certain multi-family or multi-generational households, and impossible to associate

[8] To measure the extent to which graduates receive additional education following their observation in the transcript data, we provide Appendix Table A5, which restricts our main sample to individuals observed at least two years following receipt of their PSEO degree and records ACS educational attainment information by PSEO degree level. Around 20 percent of individuals who earned a certificate degree in the PSEO report a bachelor's degree in the ACS, and 44 percent of PSEO associate graduates report a bachelor's degree. Almost half of Most Competitive bachelor's graduates report an advanced degree (compared to around 22 percent for graduates from Non-Competitive or Unrated bachelor's programs, and 8 percent for associate graduates). One might argue that this phenomenon results in misclassification of our education categories: if so, then our approach is biased *against* finding heterogeneity in the gender gap across degree levels. Nevertheless, comparisons of male postgraduation earnings across PSEO degree levels suggest that our approach produces substantial stratification: the average male graduate with an associate (certificate) degree earned $43,700 ($39,700), or roughly one-half as much as the average male graduate from a Most Competitive bachelor institution. We interpret the attainment of additional education as part of the natural school-work paths our graduates may take following initial postsecondary graduation: these paths, to the extent that they differ by gender, will be reflected in our estimates.

parents with children who are not coresidents. Using the CHCK eliminates this complication, but its time coverage prevents us from observing complete birth histories for individuals that had children at an unusually young age or earned a postsecondary degree at an unusually old age.

To avoid these censoring issues, we create a subsample of individuals who responded to the ACS during the 2016-2019 survey years and who were no older than 37 when they responded. We also impose a minimum age limit of 28 to obtain a sample of individuals observed at ages when fertility rates are relatively high. This subsample is about one-fourth as large as the main sample, but we verified that the key gender pay gap patterns were similar between the two samples.

III. Methodology

We consider two earnings concepts: total real earnings (including zeros), and log real earnings from wage and salary employment.[9] We view the two measures as complementary. The first measure does not restrict on any definition of formal employment, and thus is as inclusive as possible of differences in paid work hours. However, it may be sensitive to outlier earners or to transitory swings in self-employment earnings. The second measure excludes self-employment earnings and those without any labor-force attachment, but this results in a comparison of women and men who all participate in the traditional workforce. This measure most closely resembles the one used in canonical literature on the gender pay gap.

A. Proximate Mechanisms

We estimate a series of earnings models in terms of the following proximate mechanisms of the gender pay gap:

[9] Real earnings are expressed in US dollars at 2019 prices.

Field of degree effects. The PSEO transcripts record six-digit field codes from the Classification of Instructional Programs (CIP), which we aggregate to two digits to avoid small cells and redundant course offerings. We observe 44 distinct codes in our dataset.[10]

Occupation and industry effects. We aggregate the 535 ACS occupation codes into 316 different occupations, following the guidance of Foster et al. (2021).[11] We also aggregate ACS industry codes into a set of 18 major codes, related to those commonly used in studies of gender segregation in the labor market (e.g. Schaller at al. 2019).

Work-hours effects. To allow for non-linear relationships between work intensity and earnings, we code work hours as a flexible categorical variable rather than simply as the number of hours worked during the year. We interact six different "weeks worked" bins (1-13, 14-26, 27-39, 40-47, 48-49, 50-52) with eight different "usual hours worked per week" bins (1-20, 21-30, 31-39, 40, 41-49, 50-54, 55-64, 65-99).

We allow these mechanisms to vary flexibly across the human capital distribution by estimating separate models for each of the ten levels of degree described above. We sometimes aggregate these to a smaller number to facilitate exposition and statistical power.

B. Decomposition Framework

We start with the model

$$(1) \qquad y_{id} = \alpha_d^0 + \beta_d^0 W_{id} + \varepsilon_{id}$$

[10] Example two-digit CIP codes include Education; Engineering; Biological and Biomedical Sciences; Mathematics and Statistics; Physical Sciences; and Psychology. As a robustness check, we reran our main analyses with four-digit codes and found broadly similar results. For more details on CIP codes, see: https://nces.ed.gov/ipeds/cipcode/

[11] We thank Brad Foster for sharing the occupation crosswalk used in Foster et al. (2021), which also appears as Appendix Table A2 of that paper.

where y is an earnings outcome, W is a binary indicator taking the value 1 if individual i is a woman, and d is the given subsample (e.g. degree level).[12] When earnings are in levels, the baseline gender pay gap estimate is $\widehat{\beta_d^0}/\widehat{\alpha_d^0}$, which is the earnings difference between the average woman and average man expressed as a percentage of his earnings. When earnings are in logs, the baseline estimate is $\widehat{\beta_d^0}$, which has the standard log-point interpretation.

We then sequentially add controls for field of degree, occupation and industry, and work hours, re-estimating the model each time and tracking how the coefficient on W changes. For example, to measure the contribution of field of degree to the pay gap, we estimate

$$y_{id} = \alpha_d^1 + \beta_d^1 W_{id} + \boldsymbol{\gamma_d}' \boldsymbol{F_{id}} + \varepsilon_{id} \quad (2)$$

where $\boldsymbol{F}$ denotes the suite of two-digit CIP code fixed effects. The amount of the gender gap explained by gender differences in field of degree is given by $\widehat{\beta_d^0} - \widehat{\beta_d^1}$, while the remaining gap is $\widehat{\beta_d^1}$. Subsequent iterations yield the amount of the gap explained by occupation and industry differences conditional on field choices; the amount explained by work-hours differences conditional on occupation, industry, and field; and the residual portion left unexplained.

C. Extension of the Framework to Analyze the Child Penalty

In Section V, we utilize the CHCK subsample to perform the above decomposition for those with and those without children. The difference provides an estimate of the effects of parenthood on the gender pay gap. This treatment of parenthood is more flexible than considering it as a separate proximate mechanism. That is, parenthood is allowed to affect the gender pay gap *through*

[12] The baseline regression model controls for survey year and graduation year fixed effects as well. We also non-parametrically control for age by reweighting each degree-level-by-gender subsample so that its age distribution is equivalent to the full sample's age distribution. We provide further information in Appendix B.

its effects on field, occupation, and work-hours outcomes—without requiring a stance on the timing of fertility desires/outcomes relative to these other choices—as well as *independently* of these channels.

To illustrate, consider graduates with degree level *d*, and let *p* index parents and *n* index non-parents. We difference the gender-gap estimates from equation (1) across *p* and *n*, and then substitute terms based on equation (2):

$$\begin{aligned}(3)\qquad \hat{\pi}_d &= \hat{\beta}^0_{p,d} - \hat{\beta}^0_{n,d} \\ &= \left(\hat{\beta}^1_{p,d} + \widehat{\boldsymbol{\gamma}}'_{\boldsymbol{p,d}}\overline{\boldsymbol{DF}}_{\boldsymbol{p,d}}\right) - \left(\hat{\beta}^1_{n,d} + \widehat{\boldsymbol{\gamma}}_{\boldsymbol{n,d}}{}'\overline{\boldsymbol{DF}}_{\boldsymbol{n,d}}\right) \\ &= \left(\hat{\beta}^1_{p,d} - \hat{\beta}^1_{n,d}\right) + \left(\widehat{\boldsymbol{\gamma}}'_{\boldsymbol{p,d}}\overline{\boldsymbol{DF}}_{\boldsymbol{p,d}} - \widehat{\boldsymbol{\gamma}}_{\boldsymbol{n,d}}{}'\overline{\boldsymbol{DF}}_{\boldsymbol{n,d}}\right).\end{aligned}$$

The objects $\hat{\beta}^1_p$ and $\hat{\beta}^1_n$ are residual gender pay gap estimates by parenthood status after controlling for field; $\widehat{\boldsymbol{\gamma}}'_{\boldsymbol{p}}$ and $\widehat{\boldsymbol{\gamma}}'_{\boldsymbol{n}}$ are field fixed-effect estimates; and $\overline{\boldsymbol{DF}}_{\boldsymbol{p}}$ ($\overline{\boldsymbol{DF}}_{\boldsymbol{n}}$) is a vector of observed gender differences in the shares of parents (non-parents) choosing each field. Thus, (3) decomposes the child penalty estimate $\hat{\pi}$ into the portion attributable to the possibility that returns to male-dominated fields are higher for parents than for non-parents (or that parents' field choices were more gender-asymmetric than were non-parents'), and a residual portion. Estimating the full series of models separately for parents and non-parents, and differencing the results, constructs a decomposition of the child penalty into field, occupation, work-hours, and residual channels.[13]

IV. Main Gender Pay Gap Estimates

[13] Our approach builds on prior work to estimate the child penalty with cross-sectional data (e.g. Juhn and McCue 2017; Kleven et al. 2023). Both studies estimated quantitatively similar child penalties when they compared the results of an individual-fixed-effect model, applied to longitudinal data, to results from the cross-sectional model. These validations suggest that reasonable inferences about the child penalty can be drawn from our cross-sectional data.

Figure 1 presents gender pay gap decompositions for each of the ten levels of degree. The total height of each bar corresponds to the baseline gender pay gap estimate (i.e., after controlling for age, graduation year, and survey year effects). Green is the portion of the gap explained by gender differences in field of degree. Orange is the portion of the remaining gap explained by occupation and industry differences. Maroon is the portion of the remaining gap explained by work hours differences. Gray represents the residual portion. In a few cases, controlling for field of degree widens, rather than narrows, the gender pay gap. These cases are captured by green whiskers as opposed to opaque bars.

Although recent literature has emphasized the very top of the human capital distribution, Panel A shows a baseline pay gap that is consistent across degree levels. Panel B, which considers log wage-and-salary earnings, shows a baseline gap that generally *falls* as degree level rises. Both panels show that the remaining gap after controlling for field-of-degree sorting is smaller at higher degree levels. That is, field-of-degree sorting contributes importantly to the gender pay gap among those graduating from top bachelor's programs, but has a smaller effect on the pay gap among less-selective bachelor's, certificate, and associate graduates.

Table 3 presents decomposition estimates and standard errors to provide precise numbers behind Figure 1. For ease of exposition, we collapse the ten degree levels into six categories. We pool the two types of certificate degrees together, collapse the six bachelor's categories that are not "Most Competitive" down to three categories, and preserve the Most Competitive bachelor's category given the prior literature's focus on the top of the human capital distribution.

This table solidifies the graphical results. For total earnings, the pay gap is quite constant across degree levels; whereas for log wage and salary earnings, the pay gap is noticeably larger at the certificate level than it is at the Most Competitive bachelor's level. Field-of-degree sorting explains

25-43 percent of the gap among Most Competitive bachelor's graduates (depending on earnings measure) but is a less-important determinant of the gap at the sub-baccalaureate and less-selective bachelor's levels.[14] Occupation sorting conditional on field-of-degree sorting is a universally important determinant of the gender pay gap, explaining 30-40 percent of the gap at all degree levels. Occupation-conditional work-hours differences are an important determinant of the gap at sub-baccalaureate degree levels, explaining approximately 30 percent of the gap. In contrast, work hours trivially affects the gap among Most Competitive bachelor's graduates. In sum, approximately one-third of the *total* earnings gap remains unexplained at all degree levels. However, the unexplained portion of the *log wage-and-salary* earnings gap falls as degree level rises: only 7.7% of this gap among Most Competitive bachelor's graduates is unexplained.

A. Estimates for Full-Year Workers

In addition to considering the full population, the gender gap literature often takes interest in the subset of full-year-employed workers. Figure 2 presents gender pay gap decompositions by degree level for this group. The results are akin to what was shown above, with one exception: the work hours channel is now estimated to be small at all degree levels, regardless of earnings measure. This suggests that the larger work-hours contributions found at less-selective-bachelor's and sub-baccalaureate degree levels in the full sample are due to gender differences in the

[14] In results not presented here, we examine how our estimates change if we use four-digit as opposed to two-digit CIP codes. While residual gender pay gaps, as well as the portions explained by work hours effects, remain remarkably similar when we use four-digit CIP codes, the portion of the gender pay gap explained by field of degree increases moderately for certificate degree holders, with an offsetting decrease in the portion of the gap explained by occupation sorting. Field of degree (work hours) still affects the gender gap substantially more (less) for Most Selective bachelor's programs than for certificate programs, although the field-of-degree comparison loses statistical significance. All other significant statistical comparisons bolded in Table 2 are preserved in the alternate specification.

extensive margin of work hours. It is notable that even with this sample restriction, we still find a smaller unexplained log wage-and-salary gap at higher degree levels, regardless of whether we control for work hours.

B. Estimates for Baccalaureate Business Majors

Business is the single largest CIP code in our sample, accounting for roughly 20 percent of all bachelor's graduates.[15] Moreover, business-related occupations are often highlighted as having some of the largest gender pay gaps (e.g. Goldin 2014; Bertrand et al. 2010). Our sample affords us a unique opportunity to analyze the postgraduation earnings of Business majors across different degree levels.

Figure 3 repeats the main gender pay gap decompositions presented in Figure 1 for the subsample of Business majors with bachelor's degrees. Because all graduates in this subsample chose the same general major, we omit field-of-degree effects from the model. We find a pattern of results that is quite consistent with the findings for the full sample. Specifically, the total earnings gap is constant across Barron's ratings, with some evidence that the residual earnings gap is smallest among Business graduates from Most Competitive institutions. Moreover, the log wage-and-salary earnings gap—both with and without controls for occupation and labor supply—falls as selectivity rises. At the Most Competitive level, the log wage-and-salary earnings gap is approximately 15 log points, and it is entirely explained by occupation effects. After controlling for occupation and labor supply, women earn nearly 4 log points more than comparable men, although this difference is not statistically significant.

V. Child Penalty Analyses

[15] Business fields have their own 2-digit CIP code of 52. We also pulled Economics majors out of the Social Sciences CIP code of 45 and pooled them with the Business majors.

To keep the number of comparisons manageable and increase statistical power in our smaller CHCK subsample, we focus on two degree levels: certificate (formed by pooling all certificate degree earners together), and "Top" bachelor's (formed by pooling Highly and Most Competitive bachelor's degree earners together). Our sample size prevents us from reliably analyzing the child penalty among Business majors of different degree levels, so we return to considering the full set of fields chosen. For each degree level, we estimate the child penalty and its mechanisms according to the procedure outlined in Section III.C.

A. Main Estimates

Table 4 presents the results. Panel A shows that the total earnings gap is around 33 percentage points higher for individuals with children; Panel B shows that the log wage-and-salary earnings gap is around 45 log points higher for individuals with children. The small CHCK subsample sizes lead to a lack of precision in some of the estimates, but we still find that children significantly widen the total earnings gap through work hours and residual channels for certificate graduates, and through occupation, work hours and residual channels for Top bachelor's graduates. Moreover, the difference in the residual effects of children across degree levels is statistically significant. We find a similar pattern of effects for the log wage-and-salary earnings gap.

These results underscore the prior finding that the gender pay gap is driven by different factors at different education levels. In particular, the greater effect of children on residual channels for certificate graduates could be the result of less access to employer-provided benefits, less access to formal or informal support networks, more exposure to various forms of discrimination, or other search frictions. It could also be the result of greater children-induced work interruptions and losses of human capital for mothers (or corresponding gains for fathers), although the finding of smaller children-induced work-hours effects casts doubt on this story.

B. The Role of Children in Explaining Pay Gap Heterogeneity

Parenthood exposure varies considerably across degree levels: for example, 57.0% of certificate graduates in the CHCK subsample are parents, as compared with only 37.2% of top bachelor's graduates. Does this variation explain our findings from Section IV? What are the implications for the gender gap once the sample has completed its fertility? We close our investigation with some analyses in this regard. To motivate these analyses, note that we can map the parental-status-specific estimates of the gender pay gap (reported in Table 3) to full-sample estimates based on the parenthood rate *r*. That is, *r* times the parent decomposition + (1-*r*) times the non-parent decomposition provides an estimate of the gender pay gap decomposition for the full sample. By counterfactually varying *r*, we can project the gender pay gap under varying exposures to parenthood.

Figure 4 presents three comparisons of the log wage-and-salary gender pay gap, as estimated in the CHCK subsample, between certificate and top bachelor's graduates. First, we compare the observed gap for certificate graduates to the counterfactual gap for top bachelor's graduates if they had certificate graduates' rate of parenthood. Our first main finding from Section IV is reversed by this counterfactual: instead of the certificate gap exceeding the top bachelor's gap by around 6 log points, it is around 4 log points smaller in this counterfactual. However, we still find larger field-of-degree effects among top bachelor's graduates and larger unexplained effects among certificate graduates. Although we now find similar work-hours effects in both subsamples, occupation effects become moderately larger among top bachelor's graduates instead of being equivalent across subsamples. We find a similar pattern of results when we compare the observed pay gap among top bachelor's graduates to the counterfactual gap among certificate graduates if they had top bachelor's graduates' rate of parenthood.

The disparate rates of parenthood we observe likely stem from differences in fertility timing as opposed to differences in completed rates of ever having children (e.g. Bailey et al. 2013). Accordingly, our final comparison simulates a parenthood rate of 80% for both groups, which is a rough projection of what the national parenthood rate will be when the sample reaches ages 45-50.[16] We again match the qualitative findings of the first comparison. Overall, these findings suggest that as parenthood rates equalize across the human capital distribution, which will occur as the sample enters mature career stages, the gender pay gap will become slightly larger among the most-educated. The gap will, however, remain pronounced across the entire distribution, and the *unexplained* gap will remain largest among the least-educated.

VI. Conclusion

We leverage a unique dataset of postsecondary transcript records linked to household survey data and administrative birth records to provide a novel examination of the gender pay gap across the distribution of pre-labor-market human capital. This analysis expands a literature that has either i) provided population decompositions of the average gender pay gap; or ii) analyzed gender inequality in high-human-capital subpopulations. Our study is the first (to our knowledge) to interact these foci, breaking down a sample of postsecondary degree earners into narrowly defined human capital subgroups and analyzing the gender gap separately within each group. Our approach

[16] The median subsample member was age 33 in 2017; this person will reach age 47 in 2031. According to Table 1 of a recent report, 84.9% of women aged 45-50 in June 2020 had had at least one live birth (U.S. Census Bureau 2023). Although there has been an increase in fertility among some groups since the early stages of the pandemic (Bailey et al. 2023), the prior 13 years saw a broad-based decline in fertility (Kearney et al. 2022). It seems reasonable to expect a slight increase in the population rate of childlessness by age 45-50 over the next ten years. The 80% assumption also closely matches an extrapolation of the parenthood-age profile observed in the CHCK subsample.

extends literature on the contribution of field choice to the gender pay gap, and offers a novel decomposition of the child penalty that we use to clarify our main findings.

We present two novel sets of results. First, despite the top-heavy focus of the recent literature, the gender pay gap is not highest for top postsecondary graduates. Second, the mechanisms behind the gap vary substantially. Field-of-degree sorting explains a substantial portion of the gap among top bachelor's graduates but is a less important determinant of the gap at lower degree levels. Occupation-conditional work-hours differences are an important determinant of the gap at sub-baccalaureate degree levels, but a less-important determinant at the top. The log earnings gap among top bachelor's graduates is almost completely explained by field and occupation effects, while it is less well-explained at lower levels. The findings about field, occupation, and residual effects are robust to considering a sample of full-year-employed workers, while the findings about occupation, work-hours, and residual effects are robust to considering a sample of bachelor's graduates who majored in Business or Economics.

Our first main result is driven by the earlier fertility timing of the less-educated: if parenthood exposure were equalized across certificate and top bachelor's levels, the pay gap would become largest, albeit modestly so, at the top. The second main result is robust to fertility timing: although the child penalty is similar across certificate and top bachelor's levels, it is driven primarily by unexplained channels among certificate graduates, but almost completely by occupation and work-hours channels among top bachelor's graduates. Regardless of parental status, the gender pay gap appears to be driven mostly by Roy-style dynamics of sector and job choice among the most-educated, and to a larger extent by frictional dynamics among the less-educated. These findings suggest that broad-based gender equality cannot be achieved without a mix of policies that target the varied determinants of the gender gap across different socioeconomic positions.

TABLE 1. SUMMARY STATISTICS OF MAIN SAMPLE

Panel A. Demographics			Panel B. Human Capital and Earnings		
	Mean	SD		Mean	SD
Woman	0.59	0.49	Certificate	0.09	0.28
Age at graduation	25.2	5.0	Associate	0.31	0.46
Age	31.3	6.4	Bachelor	0.61	0.49
Graduation year	2008	3.1	Selective Bachelor	0.23	0.42
ACS year	2014	3.7	Total earnings, women	38.7	39.3
White, non-Hispanic	0.71	0.45	Total earnings, men	53.2	56.2
Black, non-Hispanic	0.09	0.28	Wage/sal earnings, women	38.0	38.6
Asian, non-Hispanic	0.05	0.22	Wage/sal earnings, men	51.5	53.9
Other, non-Hispanic	0.03	0.17	Earned < $5000, women	0.17	0.37
Hispanic	0.12	0.32	Earned < $5000, men	0.12	0.32
N = 630,000					

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.

Note: Sample membership described in Section I.A. "SD" stands for standard deviation. "Selective Bachelor" corresponds to students who graduated from an institution rated as Very Competitive, Highly Competitive, or Most Competitive according to Barron's 2004 guide. Earnings are for the 12 months prior to when the individual responded to the ACS. Earnings are inflated to 2019 prices using the CPI for all urban consumers and are expressed in thousands.

TABLE 2. AVERAGE INSTITUTIONAL CHARACTERISTICS BY BARRON'S CATEGORY

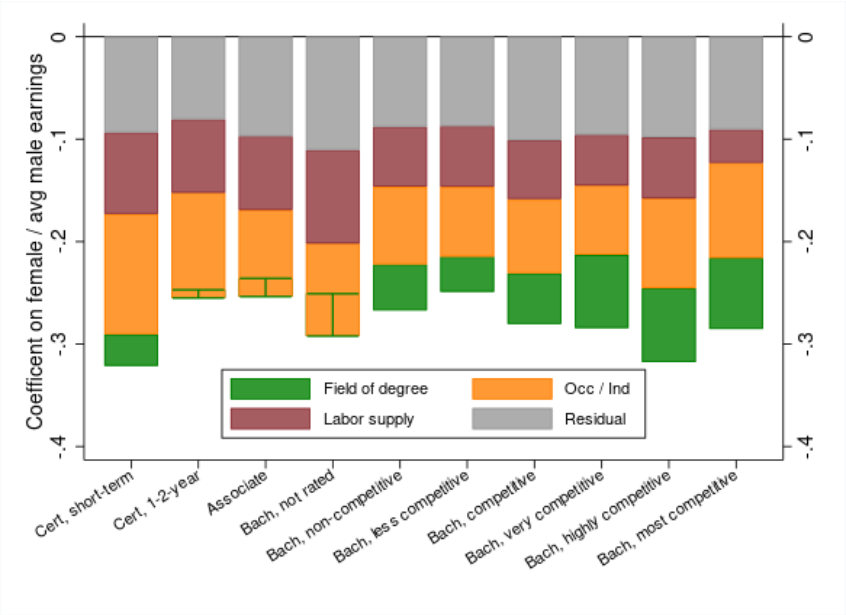
Condensed Barron's Rating	Rejection Rate	Expenditures / FTE student	Graduation Rate (4-year)	Male Total Earnings	N Students	N Institutions
Non-Competitive	0.046	4,960	0.294	52,200	36,500	25
Less Competitive	0.188	4,660	0.335	52,100	36,000	22
Competitive	0.301	5,700	0.468	55,500	150,000	64
Very Competitive	0.331	7,180	0.606	62,900	75,500	22
Highly Competitive	0.418	7,570	0.717	71,400	54,000	9
Most Competitive	0.625	10,040	0.903	85,200	12,500	6

Source: PSEO transcript records linked to the 2005-2019 ACS, 2004 Barron's Guide, and 2004-05 College Scorecard.

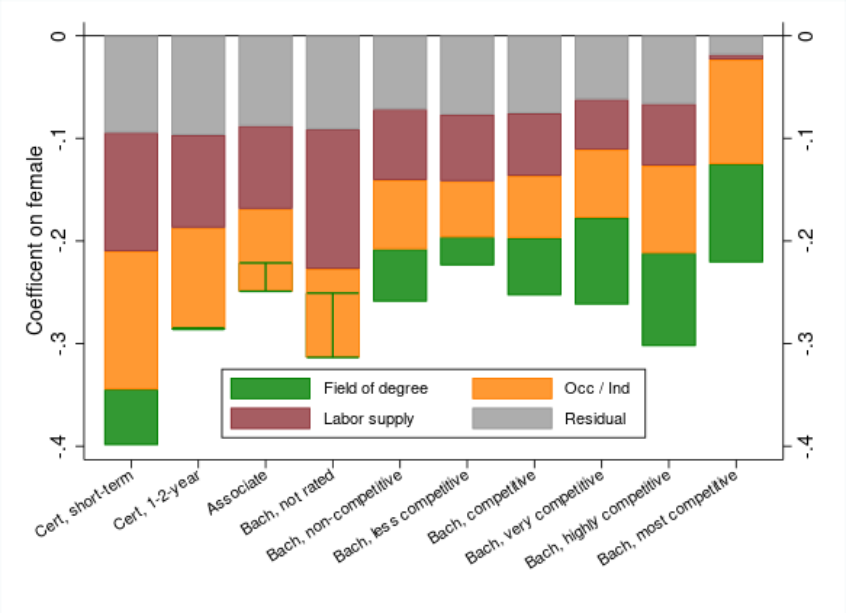
Note: Expenditures are instructional expenditures per full-time-equivalent (FTE) student, expressed in Fiscal Year 2004 dollars. Male earnings are for the 12 months prior to when the graduate responded to the ACS and are expressed at 2019 prices. Expenditures are rounded to the nearest ten, male earnings are rounded to the nearest hundred, and sample counts are rounded according to internal rounding rules for disclosure avoidance purposes.

FIGURE 1. GENDER PAY GAP DECOMPOSITIONS BY TYPE OF DEGREE, FULL SAMPLE

PANEL A. TOTAL EARNINGS



PANEL B. LOG WAGE AND SALARY EARNINGS



Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.

Note: See Section II.A for sample information. "Cert" stands for certificate degrees and "Bach" for bachelor's degrees. "Occ / Ind" stands for occupation and industry effects. Total length of each bar corresponds to the gender pay gap after controlling for age, degree year, and survey year effects. Green is the portion of the gap explained by adding field of degree effects to the pay gap regression. Orange is the portion of the remaining gap explained by adding

occupation and industry effects. Maroon is the portion of the remaining gap explained by adding fixed effects describing different levels of work hours (6 weeks worked bins interacted with 8 weekly hours bins). Gray represents the residual portion. In a few cases, controlling for field of degree widens, rather than narrows, the gender pay gap. These cases are captured by green whiskers as opposed to opaque bars.

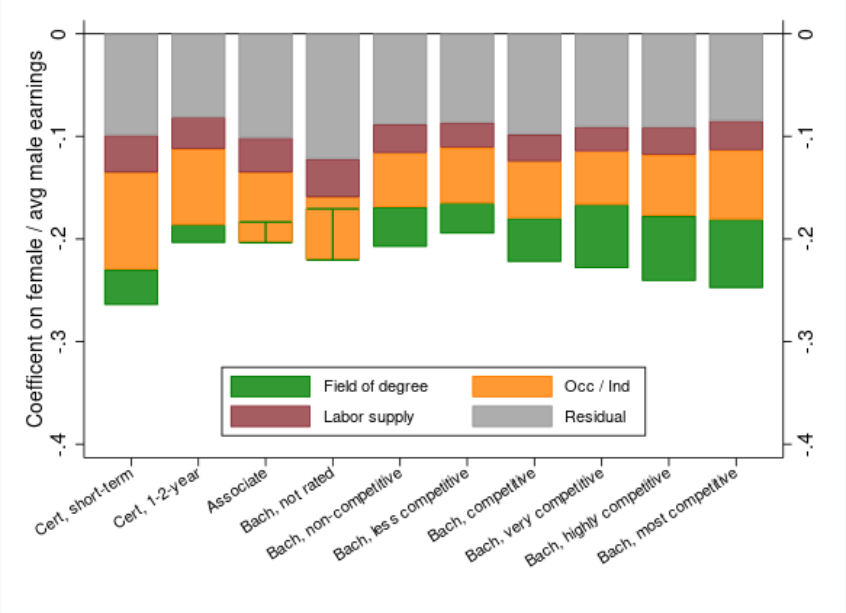
TABLE 3. GENDER PAY GAP DECOMPOSITIONS BY BINNED DEGREE LEVEL

	Non-Bachelor		Bachelor			
	Certificate	Associate	Not Selective	Selective	Very Selective	Most Selective
Panel A. Total Earnings, Full Sample						
Total gap	-0.288	**-0.236**	-0.257	-0.280	-0.300	-0.284
	(0.008)	(0.004)	(0.005)	(0.004)	(0.005)	(0.016)
Field of degree	**-0.011**	**0.018**	**-0.024**	-0.050	-0.072	-0.070
	(0.012)	(0.006)	(0.007)	(0.006)	(0.007)	(0.023)
Occupation	-0.111	-0.085	-0.077	-0.072	-0.077	-0.092
	(0.014)	(0.006)	(0.007)	(0.006)	(0.007)	(0.023)
Labor supply	**-0.076**	**-0.072**	-0.065	-0.057	-0.054	-0.032
	(0.013)	(0.006)	(0.007)	(0.006)	(0.007)	(0.021)
Residual	-0.090	-0.097	-0.091	-0.101	-0.097	-0.090
	(0.009)	(0.004)	(0.005)	(0.004)	(0.004)	(0.015)
N	54,500	192,000	90,500	150,000	130,000	12,500
Panel B. Log Wage and Salary Earnings, Full Sample						
Total gap	**-0.347**	-0.221	-0.243	-0.253	**-0.279**	-0.221
	(0.009)	(0.005)	(0.006)	(0.005)	(0.005)	(0.019)
Field of degree	**-0.027**	**0.028**	**-0.019**	-0.056	-0.087	-0.097
	(0.015)	(0.007)	(0.009)	(0.007)	(0.008)	(0.027)
Occupation	-0.119	-0.080	-0.068	-0.060	-0.075	-0.102
	(0.016)	(0.007)	(0.010)	(0.007)	(0.008)	(0.027)
Labor supply	**-0.105**	**-0.081**	**-0.079**	**-0.061**	**-0.053**	-0.005
	(0.014)	(0.006)	(0.008)	(0.006)	(0.007)	(0.023)
Residual	**-0.097**	**-0.088**	**-0.077**	**-0.076**	**-0.064**	-0.017
	(0.008)	(0.003)	(0.005)	(0.004)	(0.004)	(0.013)
N	46,000	168,000	81,500	135,000	118,000	11,000

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.

Note: Ordinary least squares models with robust standard errors appearing below coefficient estimates in parentheses. **Bold** point estimates are significantly different from the estimate for Most Selective bachelor's graduates at the 10 percent level or less.

PANEL A. TOTAL EARNINGS



PANEL B. LOG WAGE AND SALARY EARNINGS

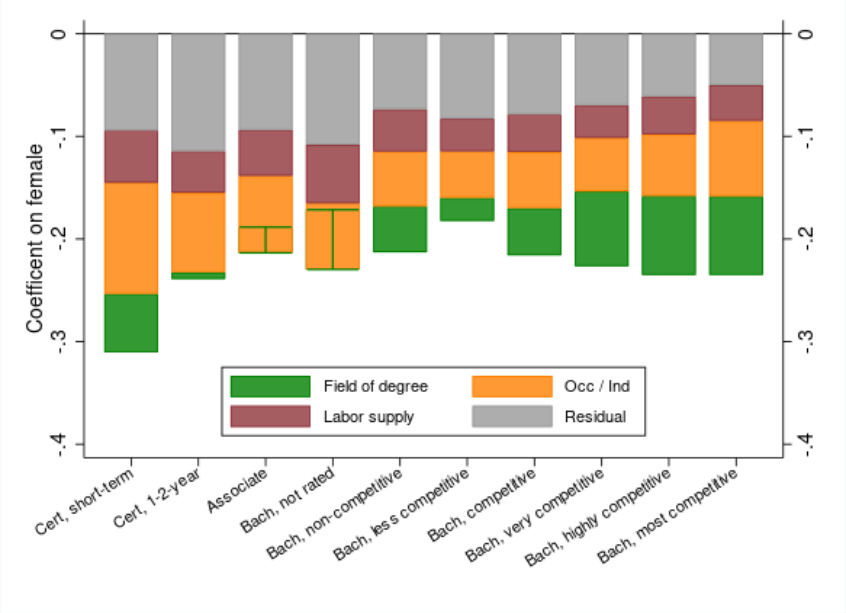
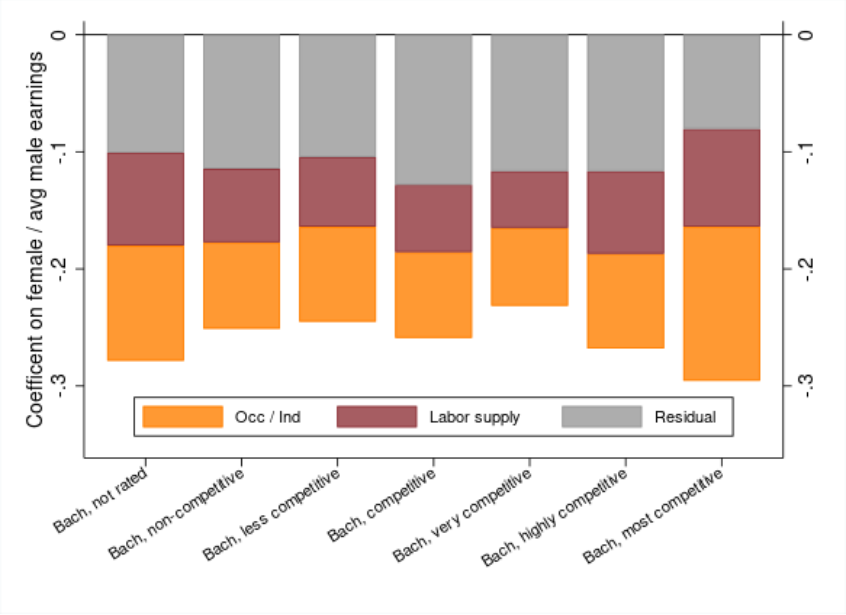


FIGURE 2. GENDER PAY GAP DECOMPOSITIONS BY TYPE OF DEGREE, YEAR-ROUND WORKERS

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.

Note: See Section II.A for sample information. Sample restricted to individuals employed for 50-52 weeks during the survey year. "Cert" stands for certificate degrees and "Bach" for bachelor's degrees. "Occ / Ind" stands for occupation and industry effects. See notes to Figure 1 for further information on the bar graphs.

PANEL A. TOTAL EARNINGS



PANEL B. LOG WAGE AND SALARY EARNINGS

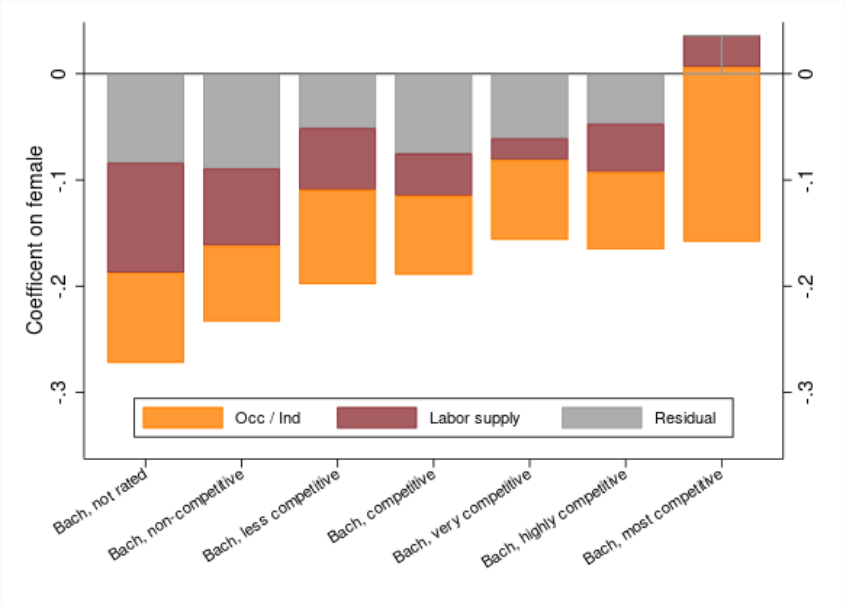


FIGURE 3. GENDER PAY GAP DECOMPOSITIONS FOR BUSINESS MAJORS BY BACHELOR'S DEGREE LEVEL

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.

Note: See Section II.A for sample information. Sample restricted to Business and Economics graduates. "Bach" refers to bachelor's degree. See notes to Figure 1 for further information on the bar graphs.

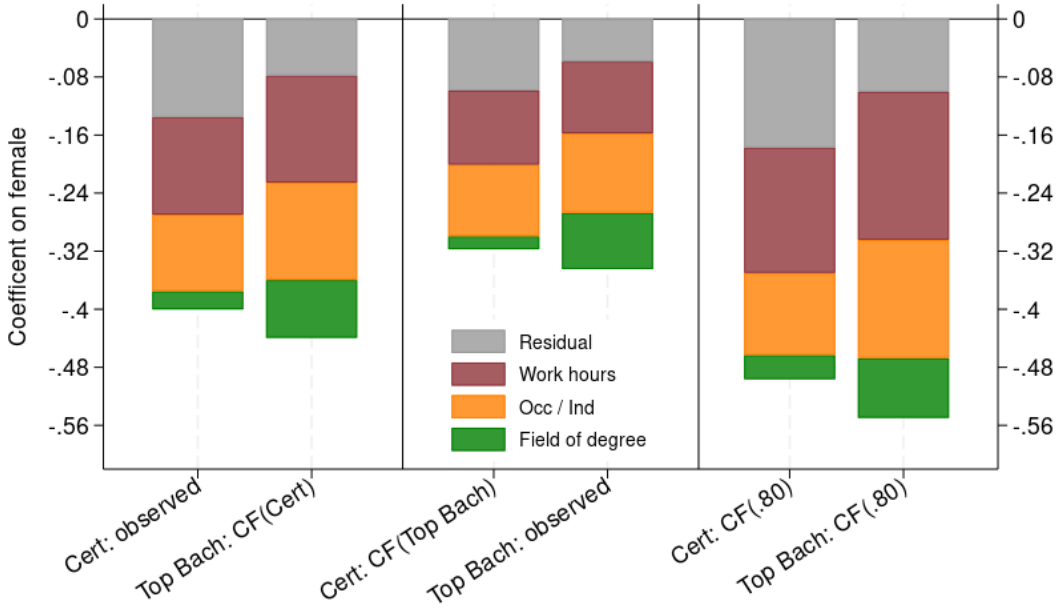
TABLE 4. GENDER PAY GAP DECOMPOSITIONS BY PARENTAL STATUS

	Certificate Grads			*Top Bachelor Grads*		
	With Children	Without Children	Difference	With Children	Without Children	Difference
Panel A. Total Earnings						
Total gap	-0.441	-0.109	**-0.332**	-0.530	-0.199	**-0.331**
	(0.019)	(0.028)	(0.034)	(0.018)	(0.014)	(0.023)
Field of degree	-0.025	-0.007	-0.018	-0.052	-0.059	0.007
	(0.032)	(0.044)	(0.054)	(0.027)	(0.021)	(0.034)
Occupation	-0.115	-0.089	-0.026	-0.163	-0.061	**-0.102**
	(0.038)	(0.048)	(0.061)	(0.028)	(0.021)	(0.035)
Labor supply	-0.125	-0.020	**-0.105**	-0.183	-0.014	**-0.169**
	(0.039)	(0.048)	(0.062)	(0.029)	(0.020)	(0.035)
Residual	-0.176	0.007	**-0.183**	-0.132	-0.065	**-0.067**
	(0.027)	(0.033)	(0.043)	(0.020)	(0.014)	(0.024)
N	6,000	4,400		7,200	13,100	
Panel B. Log Wage and Salary Earnings						
Total gap	-0.580	-0.162	**-0.418**	-0.645	-0.167	**-0.478**
	(0.024)	(0.029)	(0.038)	(0.024)	(0.015)	(0.028)
Field of degree	-0.041	-0.005	-0.036	-0.084	-0.074	-0.010
	(0.042)	(0.045)	(0.062)	(0.036)	(0.022)	(0.042)
Occupation	-0.119	-0.086	-0.033	-0.188	-0.063	**-0.125**
	(0.050)	(0.050)	(0.071)	(0.036)	(0.022)	(0.042)
Labor supply	-0.206	-0.041	**-0.165**	-0.253	-0.008	**-0.245**
	(0.045)	(0.045)	(0.064)	(0.032)	(0.019)	(0.037)
Residual	-0.214	-0.030	**-0.184**	-0.120	-0.022	**-0.098**
	(0.026)	(0.027)	(0.037)	(0.019)	(0.012)	(0.022)
N	5,100	3,900		6,200	12,400	

Source: PSEO transcript records linked to the 2005-2019 ACS, the 2004 Barron's Guide, and the Census Household Composition Key (CHCK). CHCK sample restriction described in Section II.D.

Note: "Top Bachelor" refers to graduates from Most Competitive or Highly Competitive institutions, as rated by Barron's. Ordinary least squares models with robust standard errors appearing below coefficient estimates in parentheses. Estimates in **bold** denote statistically significant effects of children on the gender pay gap or its associated component at the 10 percent level or less.

FIGURE 4. COMPARISONS OF OBSERVED AND COUNTERFACTUAL GENDER PAY GAP DECOMPOSITIONS WITH VARYING RATES OF PARENTHOOD



Source: PSEO transcript records linked to the 2005-2019 ACS, the 2004 Barron's Guide, and the Census Household Composition Key (CHCK). CHCK sample restriction described in Section II.D.

Note: These bar graphs consider the log wage-and-salary earnings outcome. "Cert" refers to certificate graduates; "Top Bach" refers to graduates from Most Competitive or Highly Competitive institutions, as rated by Barron's. "CF" refers to a counterfactual parenthood rate applied to the given subgroup, holding constant the child penalty estimate and its corresponding mechanism for that subgroup. The 80% parenthood rate applied to the last two bars represents a rough projection of the national parenthood rate for these cohorts at ages 45-50; see section V.B for further detail.

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Appendix for "The Gender Pay Gap and Its Determinants Across the Human Capital Distribution"

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A. Omitted Tables and Figures

TABLE A1. STATE HIGHER EDUCATION SYSTEMS IN OUR SAMPLE

System	PSEO		Our Sample	
	Year Coverage	2015 Share of State's Gradautes Covered	Year Coverage	N
Connecticut State Colleges and Universities (CSCU)	2001-2020	69%	2003-2013	14,500
City University of New York (CUNY)	2001-2019	44%	2003-2013	38,500
Lousiana Board of Regents (LABOR)	2001-2019	74%	2003-2013	35,500
Missouri Department of Higher Education and Workforce Development (MDHEWD)	2005-2020	50%	2005-2013	37,500
Ohio Department of Higher Education (ODHE)	2002-2020	61%	2003-2013	108,000
State Council of Higher Education for Virginia (SCHEV)	1997-2020	87%	2003-2013	93,000
State University of New York (SUNY)	2002-2019	44%	2003-2013	107,000
Texas Higher Education Coordinating Board (THECB)	2000-2019	77%	2003-2013	185,000
Utah System of Higher Education (USHE)	2010-2020	59%	2010-2013	11,500
Total				630,000

Source: PSEO transcript records linked to the 2005-2019 ACS.
Notes: Coverage statistics come from here: Post-Secondary Employment Outcomes (PSEO) (census.gov). The entries for CUNY and SUNY record the total share of New York state's graduates covered by these two systems. Sample Ns rounded for disclosure avoidance purposes.

* U.S. Census Bureau, Center for Economic Studies. Any opinions and conclusions expressed herein are those of the authors and do not reflect the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7527717: CBDRB-FY2023-CES005-020 and CBDRB-FY24-CES014-015.)
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TABLE A2. COMPARISON OF PSEO-ACS SAMPLE OF ASSOCIATE GRADUATES TO POPULATION-REPRESENTATIVE SAMPLES DRAWN FROM THE ACS

	PSEO-ACS	ACS target comparison	ACS nat'l comparison		PSEO-ACS	ACS target comparison	ACS nat'l comparison
Panel A. Demographics				Panel B. Work Hours and Earnings			
Woman	0.602	0.572	0.561	Worked FY, women	0.662	0.646	0.630
	(.001)	(.002)	(.001)		(.002)	(.002)	(.001)
White, non-Hispanic	0.734	0.679	0.632	Worked FY, men	0.748	0.745	0.735
	(.001)	(.002)	(.001)		(.002)	(.002)	(.001)
Black, non-Hispanic	0.083	0.136	0.120	Gender ratio	0.885	0.867	0.857
	(.001)	(.001)	(.001)		(.004)	(.004)	(.002)
Asian, non-Hispanic	0.028	0.012	0.046	Worked FT | FY, women	0.795	0.797	0.779
	(.000)	(.000)	(.000)		(.002)	(.002)	(.001)
Other, non-Hispanic	0.025	0.022	0.032	Worked FT | FY, men	0.894	0.903	0.900
	(.000)	(.000)	(.000)		(.002)	(.002)	(.001)
Hispanic	0.130	0.151	0.170	Gender ratio	0.889	0.883	0.866
	(.001)	(.001)	(.001)		(.003)	(.002)	(.001)
Age	31.2	31.2	32.3	Earnings, women	29.0	26.9	26.7
	(0.02)	(0.01)	(0.01)		(0.10)	(0.11)	(0.06)
				Earnings, men	41.1	41.4	41.4
					(0.17)	(0.20)	(0.10)
				Gender ratio	0.706	0.650	0.645
					(.004)	(.004)	(.002)
				W/S earnings | >0, women	33.2	31.9	32.3
					(0.10)	(0.12)	(0.06)
				W/S earnings | >0, men	44.4	45.1	45.4
					(0.17)	(0.20)	(0.10)
				Gender ratio	0.748	0.707	0.711
					(.004)	(.004)	(.002)

Source: PSEO transcript records linked to the 2005-2019 ACS; 2012-2016 ACS.

Notes: All samples in this table are restricted to individuals who have completed an associate degree only. See Section II.A of main text for PSEO-ACS sample information. We compared main summary statistics from this sample to two different samples drawn from the 2012-2016 restricted-access ACS files. (These samples contain observations from 2012-2016, with an average observation year of 2014, which matches the PSEO sample.) The "ACS target comparison" sample uses ACS base sampling weights, and is then reweighted on age to match the distribution of age observed in the PSEO sample. It also consists only of those individuals born in the 8 states whose transcript records we observe (i.e. CT, LA, MO, NY, OH, TX, UT, VA). The "ACS nat'l comparison" sample uses ACS sampling weights and considers all individuals aged 19-44. "Worked FY" is defined as being employed for at least 50 weeks out of the year, and "Worked FT" is defined as working at least 35 usual weekly hours. "W/S" stands for "wage-and-salary". The entries in the table record the sample share of individuals with the given characteristic, or the sample mean of the given continuous variable, with standard errors appearing below point estimates in parentheses. Earnings are expressed in thousands of 2019 US dollars.

TABLE A3. COMPARISON OF PSEO-ACS SAMPLE OF BACHELOR'S GRADUATES TO POPULATION-REPRESENTATIVE SAMPLES DRAWN FROM THE NSCG

	PSEO-ACS	NSCG target comparison	NSCG nat'l comparison
Panel A. Demographics and Education Attainment			
Woman	0.584	0.585	0.568
	(.001)	(.013)	(.005)
White, non-Hispanic	0.743	0.690	0.673
	(.001)	(.011)	(.004)
Black, non-Hispanic	0.076	0.103	0.076
	(.001)	(.007)	(.003)
Asian, non-Hispanic	0.059	0.054	0.107
	(.000)	(.004)	(.002)
Other, non-Hispanic	0.029	0.027	0.034
	(.000)	(.004)	(.002)
Hispanic	0.094	0.127	0.110
	(.001)	(.008)	(.003)
Age	32.3	32.3	31.3
	(0.01)	(0.17)	(.047)
Years since BA graduation	7.3	7.4	8.5
	(0.01)	(0.10)	(0.04)
BA degree only	0.705	0.741	0.708
	(.001)	(.009)	(.004)
Advanced degree	0.295	0.259	0.292
	(.001)	(.009)	(.004)
Biz/econ major for BA	0.204	0.190	0.207
	(.001)	(.012)	(.005)
STEM major for BA	0.232	0.244	0.280
	(.001)	(.008)	(.003)

	PSEO-ACS	NSCG target comparison	NSCG nat'l comparison
Panel B. Work Hours and Earnings			
Worked FT | worked, women	0.852	0.841	0.823
	(.001)	(.012)	(.005)
Worked FT | worked, men	0.925	0.924	0.917
	(.001)	(.010)	(.004)
Gender ratio	0.921	0.910	0.897
	(.001)	(.016)	(.007)
Earnings, women	48.6	48.7	48.7
	(0.12)	(1.60)	(0.56)
Earnings, men	68.5	67.0	73.6
	(0.20)	(2.92)	(1.09)
Gender ratio	0.709	0.727	0.662
	(.003)	(.040)	(.012)
Earnings | >0, women	52.8	54.7	55.1
	(0.12)	(2.27)	(0.80)
Earnings | >0, men	71.2	70.1	77.3
	(0.20)	(3.00)	(1.12)
Gender ratio	0.742	0.780	0.713
	(.003)	(.047)	(.015)

Source: PSEO transcript records linked to the 2005-2019 ACS; 2013, 2015, and 2017 National Survey of College Graduates (NSCG).

Notes: All samples in this table are restricted to individuals who have completed at least a Bachelor's degree. See Section II.A of main text for PSEO-ACS sample information. We restricted this main sample to individuals who had graduated at least two years prior to being sampled by the ACS, to match the NSCG's design of randomly sampling individuals who had responded to the ACS with a Bachelor's degree two years prior. We compared main summary statistics from this sample to two different samples drawn from the 2013, 2015, and 2017 NSCG files. The "NSCG target comparison" sample uses NSCG base sampling weights, and is then reweighted on age and years-since-graduation to match the joint distribution of both observed in the PSEO sample. It also consists only of those individuals who earned their first Bachelor's degree from a public institution in one of the 8 states whose transcript records we observe (i.e. CT, LA, MO, NY, OH, TX, UT, VA)—or from any institution in Virginia. The "NSCG nat'l comparison" sample uses NSCG sampling weights and considers all individuals aged 22-40 who earned their first BA 2-16 years ago. "Worked FT" is defined as working at least 35 usual weekly hours. The entries in the table record the sample share of individuals with the given characteristic, or the sample mean of the given continuous variable, with standard errors appearing below point estimates in parentheses. Earnings are expressed in thousands of 2019 US dollars.

TABLE A4. COMPARISON OF PSEO INSTITUTION CHARACTERISTICS TO OTHER BACHELOR'S DEGREE GRANTING INSTITUTIONS

	PSEO Institutions	All Institutions	All Public Institutions
Admission Rate	69.03	67.33	69.11
	(0.0166)	(0.0070)	(0.0078)
Average SAT Equivalent of Admitted Students	1,070	1,084	1,067
	(0.10)	(0.05)	(0.05)
Instructional Expenditures Per Full Time Equivalent Student	5,886	7,494	6,578
	(1.76)	(2.17)	(1.42)
Completion rate for first-time, full-time students	47.04	53.27	50.35
	(0.0155)	(0.0068)	(0.0075)

Source: PSEO R2021Q3 public use files and 2004-05 College Scorecard.
Note: Sample of institutions is limited to institutions that appear in both the 2004 Barron's Guide and the 2004-05 College Scorecard. Expenditures are expressed in Fiscal Year 2004 dollars. Means and standard errors are weighted based on number of undergraduate degree-seeking students enrolled at each institution.

TABLE A5. ACS EDUCATIONAL ATTAINMENT BY PSEO DEGREE LEVEL

PSEO degree level	Share with Bachelor degree	Share with advanced degree
Short-term Certificate	0.185	0.031
1-2-year Certificate	0.228	0.042
Associate	0.441	0.080
Unrated Bachelor	0.931	0.215
Non-Competitive	0.933	0.239
Less Competitive	0.929	0.223
Competitive	0.941	0.255
Very Competitive	0.952	0.308
Highly Competitive	0.952	0.338
Most Competitive	0.984	0.454

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section I.A of main text for sample information. Sample restricted to individuals observed at least two years after earning their PSEO degree, and with non-imputed ACS educational attainment information. The bachelor's degree shares for PSEO bachelor's graduates (i.e. the last seven rows of the table) should be 100 percent, but are not due to mismeasurement of educational attainment in the ACS. See Foote and Warren (2020) for more detail.

B. Further Detail on Methodology

Age Weights

Although our focus on postsecondary graduates observed 0-16 years following graduation results in a comparison of women and men at similar ages, there are multiple potential sources of age discrepancies both within and across degree levels. First, particularly at less-selective degree levels, men tend to earn their degrees at slightly older ages than women. Second, although the modal-aged certificate earner is slightly younger than the modal-aged bachelor degree earner, the average-aged certificate earner is slightly older, due to a long right tail of individuals earning their degrees in their later 20s and 30s. (The same pattern holds for associate degree earners.) Third, because of differences in data coverage across systems, and the fact that different systems have different distributions of degree levels, it is not guaranteed that the same distribution of years-since-graduation (and, consequently, age) will be observed across degree levels.

Because age is known to influence the gender pay gap, and we are interested in comparing the gap and its determinants across degree levels, it is important to ensure that the age distribution is held constant throughout our analyses. To accomplish this, we divided the full sample into six age bins: 19-23, 24-28, 29-33, 34-38, 39-43, 44-58. (Recall that the full sample consists of individuals who earned their degrees between the ages of 19 and 42, and were observed as late as 16 years following graduation.) Let s_b denote the share of the full sample belonging to bin *b*. Then, for each individual belonging to age bin *b*, gender *g,* and degree level *d*, we assign the weight

$$(1) \qquad w_b^{g,d} = \frac{s_b}{\sigma_b^{g,d}}$$

where $\sigma_b^{g,d}$ denotes the share of individuals of gender *g* and degree level *d* belonging to age bin *b*. All summary statistics and regression results reported in the paper use analytic weights *w*. These weights, in effect, set each gender-by-degree-level subsample's age distributions across the six bins equal to the distribution observed for the full sample.

Decomposition Details

Within each stratum that we consider, we estimate the total gender pay gap; the portion attributable to each explanatory variable; and the residual portion. One approach would be to estimate a single earnings regression with all covariates included and compute the standard Oaxaca-Blinder decomposition or a related variant. This approach assumes that covariates are not endogenous: observed correlation between two covariate groups X_1 and X_2 is incidental and not because X_1 causally relates to X_2. As shown by Gelbach (2016), this "full model" partials out the correlation between X_1 and X_2, and the X_1-conditional wage effect of X_2, to estimate the wage effect of X_1. This estimated partial effect is then used to compute the contribution of cross-group differences in X_1 to the observed wage gap.

We propose an alternative approach that strikes us as more appropriate for our current setting. To illustrate, let X_1 refer to field of degree and X_2 to occupation. Degree field relates to

occupation in a plausibly causal manner: if a graduate's field of degree were manipulated, one would expect a *ceteris paribus* change in their skills, preferences, and eventual occupation—and this shift in occupation *in itself* would induce a potential earnings response.[1] This suggests the estimation of a *total* effect of field of degree on earnings, without controlling for occupation, rather than the *partial* effect described above.

Accordingly, we implement a *sequential* decomposition framework, like that used in Altonji et al. (2012), to measure the gender pay gap and how it is affected by our explanatory variables of interest. We start with the baseline regression model

$$(2) \qquad y_{id} = \alpha_d^0 + \beta_d^0 W_{id} + \varepsilon_{id}$$

where y is an earnings outcome, W is a binary indicator taking the value 1 if individual i is a woman, and d is the given stratum (e.g. degree level).[2] When y corresponds to earnings in levels, the baseline gender pay gap estimate is $\widehat{\beta_d^0}/\widehat{\alpha_d^0}$, which is the earnings difference between the average woman and average man expressed as a percentage of his earnings. When y corresponds to earnings in logs, the baseline estimate is $\widehat{\beta_d^0}$, which has the standard log-point interpretation.

We then sequentially add controls for 1) field of degree, 2) occupation and industry, and 3) labor supply, re-estimating the model each time and tracking how the coefficient on W changes. For example, to measure the contribution of field of degree to the gender pay gap, we estimate

$$(3) \qquad y_{id} = \alpha_d^1 + \beta_d^1 W_{id} + \boldsymbol{\gamma_d}' \boldsymbol{F_{id}} + \varepsilon_{id}$$

where $\boldsymbol{F}$ denotes the suite of two-digit CIP code fixed effects. The amount of the gender gap explained by gender differences in field of degree is given by $\widehat{\beta_d^0} - \widehat{\beta_d^1}$, while the remaining gap is $\widehat{\beta_d^1}$. Subsequent iterations yield the amount of the gap explained by occupation and industry differences conditional on field choices; the amount explained by labor supply differences conditional on occupation, industry, and field; and the residual portion left unexplained.

While the ordering of field-of-degree choices before occupation and labor supply is "natural", it is worth elaborating on our decision to control for occupation ahead of labor supply. One might argue that individuals set their occupation and labor supply choices simultaneously, or that labor supply preferences are independent of occupation-specific preferences and skills. In this case, one should not expect an intervention that causes a change in graduate i's occupation choice to have spillover effects on i's labor supply. Such a process is more consistent with first

[1] If eventual field and occupation outcomes were predetermined on entry to college by a set of other factors, then it would be wrong to suspect such a dependence. However, the literature suggests that student experiences in college are critical for the choice of major, as they receive information about their own preferences and skills through coursework and extracurricular activities, and update their priors (e.g. Zafar 2011; Stinebrickner and Stinebrickner 2014; Kahn and Ginther 2017; Patnaik et al. 2020). Moreover, different fields confer different skills that are relevant for earnings outcomes (e.g. Arcidiacono 2004). Work by Calkins et al. (2023) finds that a large set of baseline characteristics, including *intent* to major in STEM as a freshman, explains only one quarter of observed variation in the decision to complete a STEM degree.

[2] The baseline regression model controls for a suite of age, survey year, and graduation year fixed effects as well, which are absorbed into the constant term when the decompositions are computed.

controlling for field of degree, and then estimating a decomposition on the residual gap that controls simultaneously for occupation and labor supply. Our decision to proceed sequentially is consistent with assumptions that labor supply is one of a correlated set of considerations that drive occupation choice. It also allows our study to engage with the popular hypothesis that the sorting of men and women into higher- versus lower-hours positions is an important mechanism of the gender gap within skilled professional occupations (Goldin 2014).

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